Volatility Trading

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Our Goal

- ATM SPX options (13 years of real market data)
- Strategy based on implied volatility (IV)
- IV term structure prediction
- Build a trading rule based on this signal

Quantitative Framework (1/2)

- **Put–Call parity $\rightarrow$ Fair Rate r**

$$r = \frac{1}{\tau} \ln\left(\frac{S - K}{C_t - P_t}\right)$$

- **Spot $\times e^{-rt}$ $\rightarrow$ Forward / Log-moneyness**

$$F_T = S_t \, e^{r\tau}$$

- **PCA model $\rightarrow$ Estimate Term Structure**

$$v_t \equiv \begin{pmatrix} \mathrm{IV}^2_{\mathrm{ATM}}(T_1, t) \\ \mathrm{IV}^2_{\mathrm{ATM}}(T_2, t) \\ \mathrm{IV}^2_{\mathrm{ATM}}(T_3, t) \\ \mathrm{IV}^2_{\mathrm{ATM}}(T_4, t) \\ \mathrm{IV}^2_{\mathrm{ATM}}(T_5, t) \end{pmatrix} \approx \mu + \mathbf{B}\,\mathbf{f}_t, \qquad \mathbf{B} \in \mathbb{R}^{5\times 3},\ \mathbf{f}_t \in \mathbb{R}^3.$$

Quantitative Framework (2/2)

- **PCA**

$$v_{i,t} \approx \mu_i + \beta_{i,1} f_{1,t} + \beta_{i,2} f_{2,t} + \beta_{i,3} f_{3,t}, \qquad i = 1, \ldots, 5,$$

where $\beta_{i,j}$ are elements of **B**.

- **AR(1)$\rightarrow$ Predict t+1 Values**

$$f_{1,t+1} = \phi_1 f_{1,t}$$

$$f_{2,t+1} = \phi_2 f_{2,t}$$

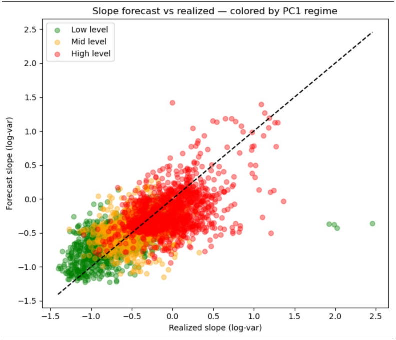
$$f_{3,t+1} = \phi_3 f_{3,t}$$

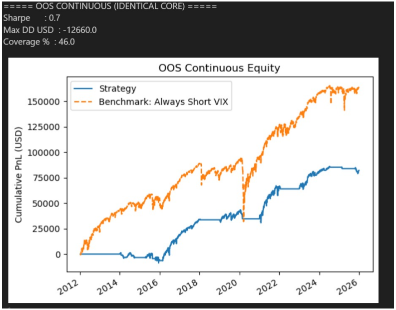
VIX and Calendar Spread

- What is a VIX strategy?
  - Short VIX in contango regimes
  - Avoid high PC1 regime
  - Stop strategy if volatility of a volatility (volga) is high
- What is a calender spread strategy?
  - Buy long-dated option (60 days)
  - Sell short-dated option (30 days)
  - Same strike price (ATM)

  - Trading rule based on slope forecast of IV term structure
  - Profit from time decay (theta), and volatility sensitity (vega)

VIX Strategy Results





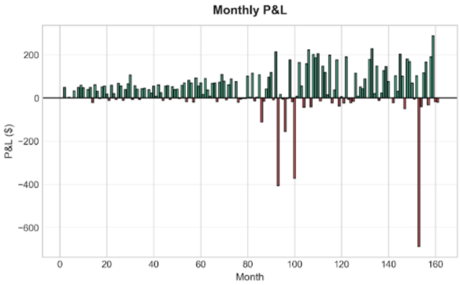
Strategy results

- Sharpe: 0.7
- Max DD: -12660
- Coverage: 46%

Benchmark results

- Sharpe: 0.637
- Max DD: -62250
- Coverage: 100%

Calendar Spread Strategy Results



PERFORMANCE SUMMARY

Total Return:		6.51%
Sharpe Ratio:		1.23
Max Drawdown:		-0.78%
Win Rate:		39.7%
Total Trades:		1031
Transaction Costs:	$	2,597

Validation

- VIX: Rolling window
- Avoid look ahead bias
- Calendar Spread: PBO = 50% -> high overfitting risk
- Both: Negative gamma loses money high vol regimes (COVID crash, tariff rally)